1.(a). $J(\theta) = \frac{1}{2}\sum_{i=1}^{m}(\theta^T x^{(i)} - y^{(i)})^2$

$$H_{jk} = \frac{\partial^2 J(\theta)}{\partial\theta_j \partial\theta_k}$$

$$\frac{\partial J(\theta)}{\partial\theta_j} = \sum_{i=1}^{m}(\theta^T x^{(i)} - y^{(i)}) \cdot x_j^{(i)}$$

$$\frac{\partial^2 J(\theta)}{\partial\theta_j \partial\theta_k} = \frac{\partial}{\partial\theta_k}\left(\frac{\partial J(\theta)}{\partial\theta_j}\right) = \sum_{i=1}^{m} x_j^{(i)} x_k^{(i)}$$

Namely, $H_{jk} = \sum_{i=1}^{m} x_j^{(i)} x_k^{(i)}$.

Check out the notation: $x_j^{(i)}$ refers to the j-th feature of the i-th observation.

We can find out that: $H = X^T X$ (X is the design matrix)

(b). Based on Newton's method in the multidimensional setting:

If you can not figure out the derivation of (*), think about this: since we know $\theta' = \theta^*$, what should it be? $(X^T X)^{-1} X^T y = \theta - (X^T X)^{-1} \vec{\beta}$

$$(X^T X)^{-1}(X^T y + \beta) = \theta$$

$$X^T y + \beta = X^T X \theta$$

$$\beta = X^T X \theta - X^T y$$

$$\beta = X^T (X\theta - y)$$

$$\theta = \theta - H^{-1} \nabla_\theta l(\theta).$$

$$\nabla_\theta l(\theta) = \begin{bmatrix} \sum_{i=1}^{m}(\theta^T x^{(i)} - y^{(i)}) \cdot x_1^{(i)} \\ \sum_{i=1}^{m}(\theta^T x^{(i)} - y^{(i)})\, x_2^{(i)} \\ \vdots \\ \sum_{i=1}^{m}(\theta^T x^{(i)} - y^{(i)})\, x_n^{(i)} \end{bmatrix} \overset{(*)}{=} X^T(X\theta - \vec{y})$$

After one iteration: $\theta' = \theta - (X^T X)^{-1} \nabla_\theta l(\theta)$

$$= \theta - (X^T X)^{-1} X^T (X\theta - \vec{y})$$

$$= \theta - X^{-1}(X^T)^{-1} X^T X \theta + (X^T X)^{-1} X^T \vec{y}$$

$$= \theta - \theta + (X^T X)^{-1} X^T \vec{y} = \theta^*$$

Namely, $\theta' = \theta^*$ #.

3. (a). $J(\Theta) = \frac{1}{2}\sum_i \sum_j \left((\Theta^T x^{(i)})_j - y_j^{(i)}\right)^2$

$$= \frac{1}{2}\sum_i \sum_j (X\Theta - Y)_{ij}^2$$

$$= \frac{1}{2}\sum_i \left((X\Theta - Y)^T (X\Theta - Y)\right)_{ii}$$

$$= \frac{1}{2} tr\left((X\Theta - Y)^T (X\Theta - Y)\right).$$

(b). $\nabla_\Theta J(\Theta) = \nabla_\Theta \left[\frac{1}{2} tr\left((X\Theta - Y)^T (X\Theta - Y)\right)\right]$

$$= \nabla_\Theta \left[\frac{1}{2} tr\left(\Theta^T X^T X\Theta - \Theta^T X^T Y - Y^T X\Theta - Y^T Y\right)\right]$$

$$= \frac{1}{2}\nabla_\Theta \left[tr(\Theta^T X^T X\Theta) - tr(\Theta^T X^T Y) - tr(Y^T X\Theta) + tr(Y^T Y)\right]$$

$$= \frac{1}{2}\nabla_\Theta \left[tr(\Theta^T X^T X\Theta) - 2tr(Y^T X\Theta) + tr(Y^T Y)\right]$$

Considering these three equations: ① $\nabla_\Theta (tr(Y^T Y)) = 0$

② $\nabla_{A^T} ABA^T C = B^T A^T C^T + BA^T C$ (let $A^T = \Theta$, $B = X^T X$, $C = I$)

③ $\nabla_A tr AB = B^T$. ($\nabla_\Theta tr(Y^T X\Theta) = (\nabla_{\Theta^T} tr(Y^T X\Theta))^T = (\nabla_{\Theta^T}(\Theta^T X^T Y))^T$

$= ((X^T Y)^T)^T = X^T Y$.

$$\Rightarrow \nabla_\Theta J(\Theta) = \frac{1}{2}\left[X^T X\Theta + X^T X\Theta - 2X^T Y\right]$$

$$= X^T X\Theta - X^T Y.$$

By setting this expression to zero, we have:

$$\Theta = (X^T X)^{-1} X^T Y.$$

(c). $\vec{y_j} = \begin{bmatrix} y_j^{(1)} \\ \vdots \\ y_j^{(m)} \end{bmatrix}$

$$\Rightarrow \theta_j = (X^T X)^{-1} X^T \vec{y_j}$$

If we line up our θ_j ($j = 1, 2, \cdots p$):

$$[\theta_1, \theta_2, \cdots, \theta_p] = [(X^T X)^{-1} X^T \vec{y_1} \cdots (X^T X)^{-1} X^T \vec{y_p}]$$

$$= (X^T X)^{-1} X^T [\vec{y_1} \cdots \vec{y_p}]$$

$$= (X^T X)^{-1} X Y = \Theta$$

Namely, p individual OLS gives the exact same solution as the multivariate least squares.

4. (a) $\ell(\varphi) = \log \prod_{i=1}^{m} p(x^{(i)}, y^{(i)}; \varphi)$

$$= \log \prod_{i=1}^{m} p(x^{(i)} | y^{(i)}; \varphi) \cdot p(y^{(i)}; \varphi)$$

$$= \log \prod_{i=1}^{m} \left(\prod_{j=1}^{n} p(x_j^{(i)} | y^{(i)}; \varphi) \right) p(y^{(i)}; \varphi)$$

$$= \sum_{i=1}^{m} \left(\log p(y^{(i)}; \varphi) + \sum_{j=1}^{n} \log p(x_j^{(i)} | y^{(i)}; \varphi) \right)$$

$$= \sum_{i=1}^{m} \Big[y^{(i)} \log \phi_y + (1-y^{(i)}) \log (1-\phi_y)$$
$$+ \sum_{j=1}^{n} \left(x_j^{(i)} \log \phi_{j|y^{(i)}} + (1-x_j^{(i)}) \log (1-\phi_{j|y^{(i)}}) \right) \Big]$$

(b). The only terms in $\ell(\varphi)$ which have non-zero gradient with respect to $\phi_{j|y=0}$ are those which include $\phi_{j|y^{(i)}}$

$$\Rightarrow \nabla_{\phi_{j|y=0}} \ell(\varphi) = \nabla_{\phi_{j|y=0}} \sum_{i=1}^{m} \left(x_j^{(i)} \log \phi_{j|y^{(i)}} + (1-x_j^{(i)}) \log (1-\phi_{j|y^{(i)}}) \right)$$

$$= \nabla_{\phi_{j|y=0}} \sum_{i=1}^{m} \Big(x_j^{(i)} \log \phi_{j|y=0} \, 1\{y^{(i)}=0\}$$
$$+ (1-x_j^{(i)}) \log (1-\phi_{j|y=0}) \, 1\{y^{(i)}=0\} \Big)$$

$$= \sum_{i=1}^{m} \left(x_j^{(i)} \frac{1}{\phi_{j|y=0}} 1\{y^{(i)}=0\} - (1-x_j^{(i)}) \frac{1}{1-\phi_{j|y=0}} 1\{y^{(i)}=0\} \right).$$

Setting $\nabla_{\phi_{j|y=0}} \ell(\varphi) = 0$ gives:

$$0 = \sum_{i=1}^{m} \left(x_j^{(i)} \frac{1}{\phi_{j|y=0}} 1\{y^{(i)}=0\} - (1-x_j^{(i)}) \frac{1}{1-\phi_{j|y=0}} 1\{y^{(i)}=0\} \right).$$

times $\phi(1-\phi)$

$$\sum_{i=1}^{m} \left(x_j^{(i)} (1-\phi_{j|y=0}) 1\{y^{(i)}=0\} - (1-x_j^{(i)}) \phi_{j|y=0} 1\{y^{(i)}=0\} \right)$$

$$= \sum_{i=1}^{m} \left((x_j^{(i)} - \phi_{j|y=0}) 1\{y^{(i)}=0\} \right)$$

$$= \sum_{i=1}^{m} \left(x_j^{(i)} \cdot 1\{y^{(i)}=0\} \right) - \phi_{j|y=0} \sum_{i=1}^{m} 1\{y^{(i)}=0\}$$

$$= \sum_{i=1}^{m} \left(1\{x_j^{(i)}=1 \wedge y^{(i)}=0\} \right) - \phi_{j|y=0} \sum_{i=1}^{m} 1\{y^{(i)}=0\}.$$

$$\Rightarrow \hat{\phi}_{j|y=0} = \frac{\sum_{i=1}^{m} 1\{x_j^{(i)}=1 \wedge y^{(i)}=0\}}{\sum_{i=1}^{m} 1\{y^{(i)}=0\}}$$

$\phi_{j|y=1}$ proceeds in the identical manner.

To solve for ϕ_y,

$$\nabla_{\phi_y} l(\varphi) = \nabla_{\phi_y} \sum_{i=1}^{m} \left(y^{(i)} \log \phi_y + (1-y^{(i)}) \log (1-\phi_y) \right)$$

$$= \sum_{i=1}^{m} \left(y^{(i)} \frac{1}{\phi_y} - (1-y^{(i)}) \frac{1}{1-\phi_y} \right)$$

Then setting $\nabla_{\phi_y} = 0$:

$$0 = \sum_{i=1}^{m} \left(y^{(i)} \frac{1}{\phi_y} - (1-y^{(i)}) \frac{1}{1-\phi_y} \right)$$

$$= \sum_{i=1}^{m} \left(y^{(i)} (1-\phi_y) - (1-y^{(i)}) \phi_y \right)$$

$$= \sum_{i=1}^{m} y^{(i)} - \sum_{i=1}^{m} \phi_y$$

$$\Rightarrow \phi_y = \frac{\sum_{i=1}^{m} 1\{y^{(i)}=1\}}{m}$$

(c).

$$p(y=1 \mid x) \geq p(y=0 \mid x)$$

$$\Leftrightarrow \frac{p(y=1 \mid x)}{p(y=0 \mid x)} \geq 1.$$

$$\Leftrightarrow \frac{\frac{p(y=1) \cdot p(x \mid y=1)}{p(x)}}{\frac{p(y=0)\, p(x \mid y=0)}{p(x)}} \geq 1$$

$$\Leftrightarrow \frac{\left(\prod_{j=1}^{n} p(x_j \mid y=1) \right) \cdot p(y=1)}{\left(\prod_{j=1}^{n} p(x_j \mid y=0) \right) p(y=0)} \geq 1$$

$$\Leftrightarrow \frac{\left(\prod_{j=1}^{n} (\phi_{j|y=1})^{x_j} (1-\phi_{j|y=1})^{1-x_j} \right) \phi_y}{\left(\prod_{j=1}^{n} (\phi_{j|y=0})^{x_j} (1-\phi_{j|y=0})^{1-x_j} \right) (1-\phi_y)} \geq 1$$

$$\Leftrightarrow \sum_{j=1}^{n} \left(x_j \log\left(\frac{\phi_{j|y=1}}{\phi_{j|y=0}}\right) + (1-x_j) \log\left(\frac{1-\phi_{j|y=1}}{1-\phi_{j|y=0}}\right) \right) + \log\left(\frac{\phi_y}{1-\phi_y}\right) \geq 0$$

$$\Leftrightarrow \sum_{j=1}^{n} x_j \log\left(\frac{\phi_{j|y=1} \cdot (1-\phi_{j|y=0})}{\phi_{j|y=0} \, (1-\phi_{j|y=1})}\right) + \sum_{j=1}^{n} \log\left(\frac{1-\phi_{j|y=1}}{1-\phi_{j|y=0}}\right) + \log\left(\frac{\phi_y}{1-\phi_y}\right) \geq 0$$

$$\Leftrightarrow \theta^T \begin{bmatrix} 1 \\ x \end{bmatrix} \geq 0$$

where $\theta_0 = \sum_{j=1}^{n} \log\left(\frac{1-\phi_{j|y=1}}{1-\phi_{j|y=0}}\right) + \log\left(\frac{\phi_y}{1-\phi_y}\right)$

$$\theta_j = \log \frac{\phi_{j|y=1} (1-\phi_{j|y=0})}{\phi_{j|y=0} (1-\phi_{j|y=1})} \quad j = 1, 2, \cdots, n.$$

5. (a). $p(y;\phi) = (1-\phi)^{y-1}\phi$

$$= \exp[\log(1-\phi)^{y-1} + \log\phi]$$

$$= \exp[(y-1)\log(1-\phi) + \log\phi]$$

$$= \exp[y\log(1-\phi) - \log(\frac{1-\phi}{\phi})]$$

$b(y)=1 \quad \eta = \log(1-\phi) \quad T(y) = y \quad a(\eta) = \log(\frac{1-\phi}{\phi}) = \log(\frac{e^\eta}{1-e^\eta})$.

(b). $g(\eta) = E[y;\eta] = \frac{1}{\phi} = \frac{1}{1-e^\eta}$

(c). $l_i(\theta) = \log[\exp(\theta^T x^{(i)} \cdot y^{(i)} - \log(\frac{e^{\theta^T x^{(i)}}}{1-e^{\theta^T x^{(i)}}}))]$

$$= \log[\exp(\theta^T x^{(i)} - \log(\frac{1}{e^{-\theta^T x^{(i)}}-1}))]$$

$$= \theta^T x^{(i)} y^{(i)} + \log(e^{-\theta^T x^{(i)}} - 1)$$

$$\frac{\partial}{\partial\theta_j} l_i(\theta) = x_j^{(i)} y^{(i)} + \frac{e^{-\theta^T x^{(i)}}}{e^{-\theta^T x^{(i)}} - 1} \cdot (-x_j^{(i)})$$

$$= x_j^{(i)} y^{(i)} - \frac{1}{1-e^{-\theta^T x^{(i)}}} x_j^{(i)}$$

$$= (y^{(i)} - \frac{1}{1-e^{-\theta^T x^{(i)}}}) x_j^{(i)}$$

Update rule: $\theta_j := \theta_j + \alpha \frac{\partial l_i(\theta)}{\partial\theta_j}$

Namely, $\theta_j := \theta_j + \alpha(y^{(i)} - \frac{1}{1-e^{\theta^T x^{(i)}}}) x_j^{(i)}$